The torque of F about O' is,

$$\overrightarrow{O'P}\times\vec{F}=\left(\overrightarrow{O'O}+\overrightarrow{OP}\right)\times\vec{F}$$
$$=\left(\overrightarrow{O'O}\times\vec{F}\right)+\left(\overrightarrow{OP}\times\vec{F}\right)$$

As $\overrightarrow{O'O}\times\vec{F}$ is perpendicular to $\overrightarrow{O'O}$, this term will not have a component along AB. Thus, the component of $\overrightarrow{O'P}\times\vec{F}$ is equal to that of $\overrightarrow{OP}\times\vec{F}$.

5.2.3 Torque and Angular Acceleration

Let us consider a rigid body rotating about a fixed axis. A point mass m in the body will execute a circular motion about a fixed axis as shown in Figure 5.11. A tangential force $\vec{F}$ acting on the point mass produces the necessary torque for this rotation. This force $\vec{F}$ is perpendicular to the position vector $\vec{r}$ of the point mass.

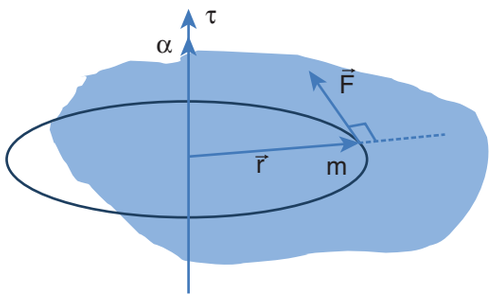


Figure 5.11 Torque and Angular acceleration

The torque produced by the force on the point mass m about the axis can be written as,

$$\tau=rF\sin90^\circ=rF \quad \left[\because \sin90^\circ=1\right]$$
$$\tau=rma \quad \left[\because \left(F=ma\right)\right]$$
$$\tau=rmr\alpha=mr^2\alpha \quad \left[\because \left(a=r\alpha\right)\right]$$

$$\tau=\left(mr^2\right)\alpha \qquad (5.14)$$

Hence, the torque of the force acting on the point mass produces an angular acceleration (α) in the point mass about the axis of rotation.

In vector notation,

$$\vec{\tau}=\left(mr^2\right)\vec{\alpha} \qquad (5.15)$$

The directions of τ and α are along the axis of rotation. If the direction of τ is in the direction of α, it produces angular acceleration. On the other hand if, τ is opposite to α, angular deceleration or retardation is produced on the point mass.

The term mr^2 in equations 5.14 and 5.15 is called moment of inertia (I) of the point mass. A rigid body is made up of many such point masses. Hence, the moment of inertia of a rigid body is the sum of moments of inertia of all such individual point masses that constitute the body $\left(I=\sum m_i r_i^2\right)$. Hence, torque for the rigid body can be written as,

$$\vec{\tau}=\left(\sum m_i r_i^2\right)\vec{\alpha} \qquad (5.16)$$
$$\vec{\tau}=I\vec{\alpha} \qquad (5.17)$$

We will learn more about the moment of inertia and its significance for bodies with different shapes in section 5.4.

5.2.4 Angular Momentum

The angular momentum in rotational motion is equivalent to linear momentum in translational motion. The angular

UNIT-5(XI-Physics_Vol-1).indd 224 06-01-2022 17:46:45